→ Multiplicative inverse

$a \times \frac{1}{a} = 1$

Eg = $5 \times \frac{1}{5} = 1$

↓

Multiplicative Inverse

(Note:-)

1) For subtraction and divison all the properties i.e. commutativity, associtivity will not follow.

2) If we Add, Subtract, divide, multiply 2 irrationals, result may be rational or irrational

→ Addition

- $\text{Ir} + \text{Ir} = \text{Ir} / r$

Eg → $\sqrt{3} + \sqrt{3} = 2\sqrt{3}$ (Irrational)

- $(2+\sqrt{3}) + (2-\sqrt{3}) = 4$

→ subtraction

- $\sqrt{3} - \sqrt{3} = 0$ (rational)
- $3\sqrt{5} - 2\sqrt{5} = \sqrt{5}$ (Irrational)

→ Multiplication

- $\sqrt{5} \times \sqrt{5} = 5$ (rational)
  Ir Ir
- $\sqrt{5} \times \sqrt{3} = \sqrt{15}$ (irrational)

→ Divison

⇒ $2\sqrt{5} \div \sqrt{5} = 2$ (rational)

- $\sqrt{5} \div \sqrt{3} = \frac{\sqrt{5}}{\sqrt{3}}$ (irrational)

Note -

1. The sum or difference of rational or irrational is irrational.
   - $r + \text{Ir} = \text{Ir}$
   - $r - \text{Ir} = \text{Ir}$

Eg → $2 + \sqrt{3}$ → Irrational
$5 - \sqrt{3}$ → Irrational

2. The product of or quotient of non-zero rational number and irrational no. is always irrational.
   $r \times \text{Ir} = \text{Ir}$

Eg → $2 \times \sqrt{3} = 2\sqrt{3}$

↓

Irrational

if rational = 0

∴ $0 \times \sqrt{3} = 0$ (Rational)

Date

Radicals.

$\sqrt[n]{x}$ → Radical expression

x → Radicand

n → Index / degree of radical / order

n

$n>0$	$n=0$	$n<0$
n is ⊕		n is ⊖
↓		complex no.
class 10th	Eg → $\sqrt[0]{0} = 0^{1/0}$ → (N.D)	(Radical value)
n is always	$\sqrt[-1]{0} = (0)^{-\frac{1}{2}} = \frac{1}{0^{1/2}}$ (N.D)	
Real no.		
Except '0'	$\sqrt[2]{0} = (0)^{1/2} = 0$	Eg → $\sqrt[2]{-5} = (-5)^{1/2}$
⇒ n can have any value except '0'	n can have only ⊕ value	$\sqrt[3]{-16} = (-16)^{1/3} = 1$
		$\sqrt[0]{-25} = (-25)^{1/0}$ → (N.D) $(16)^{1/3}$
Eg → $5\sqrt{3} = 3^{1/5}$	Eg →	n can have real value except '0'
$\sqrt[-2]{3} = (3)^{-\frac{1}{2}} = \frac{1}{3^{1/2}}$		
$\sqrt[0]{5} = (5)^{1/0} = N.D$		

- $\sqrt{\ } = (\ )^{1/2}$
- $\sqrt[3]{\ } = (\ )^{1/3}$
- $\sqrt[5]{\ } = (\ )^{1/5}$
- $\sqrt[4]{\ } = (\ )^{1/4}$
- $\sqrt[n]{x} = (x)^{1/n}$

Date

Surds

$\sqrt[n]{a}$

here, a is positive rational no.

n → positive integer / natural no.

⇒ $n > 1$

i.e. $n = 2, 3, 4, 5 \ldots$

- If 'a' is the positive rational number which is not the nth power (n is a natural no./positive integer), of any rational number then the irrationals numbers are called surds.

Eg → $\sqrt[2]{\pi}$ = not a surd
↓ irrational

- $\sqrt{2+\sqrt{3}}$ → not a surd
↓ irrational

- $\sqrt{9} = 3$ (not a surd)

- $\sqrt[3]{4}$ = as it is → surd
- $\sqrt[3]{8} = (2^3)^{\frac{1}{3}} = 2$ → (not a surd)
- $\sqrt{25} = 5$ (not a surd)
- $\sqrt{3}$ = (surd)
- $\sqrt[4]{625} = (5^4)^{1/4} = 5$ (not a surd)

Note: → Every surd is an irrational no. but every irrational no. is not a surd.

- π is an irrational no. but not a surd.

Spiral

Types of surds.

1.) simple / monomial / pure surd.
Eg → $\sqrt{2}$, $\sqrt[3]{3}$ ... etc.

2.) Binomial surd - combination of 2 surds.
eg - $\cancel{\sqrt{2}+\sqrt{5}}$ $\sqrt{3}+\sqrt{5}$, $\sqrt[3]{2}-\sqrt{5}$ .... etc.

3.) Mixed surd :
Eg → $2+\sqrt{3}$

4.) compound surd. → more than two surds there.
Eg → $\sqrt{2}+\sqrt{3}-\sqrt{5}$ etc.

5.) similar surds - $\underbrace{\sqrt{2} \text{ and } 3\sqrt{2}}_{\text{same}}$ are similar

Eg → $3\sqrt{7}$ and $4\sqrt{7}$ are sameimilar

6.) Quadratic surd - (square root)
$\sqrt{2}, \sqrt{3}, \sqrt{5}$ etc.

• If surds are similar
• Addition -
Eg → $2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$
• $2\sqrt{5} + 3\sqrt{7}$ = can't add → They are not similar surd

• $5\sqrt{3} + \sqrt{27}$
$5\sqrt{3} + 3\sqrt{3}$
$= 8\sqrt{3}$

• Subtraction
Eg. $3\sqrt{2} - 5\sqrt{2} = -2\sqrt{2}$
• $\Rightarrow 7\sqrt{3} - 2\sqrt{3} - 5\sqrt{3} = 0$
• $\Rightarrow 5\sqrt{2} - 24\sqrt{2} = -19\sqrt{2}$

• Multiplication and divison of surds
→ Order should be same
Eg → $2\sqrt{3} \times 5\sqrt{3} = 10 \times 3 = 30$

Eg → $\underbrace{2\sqrt{3} \times \sqrt[4]{2}}_{\text{Not similar}} = \times$ (N.P)

$2\sqrt{3} \times \sqrt[4]{2}$
$2(3)^{\frac{1}{2}\times\frac{2}{2}} \times (2)^{1/4}$
$2(3)^{2/4} \times (2)^{1/4}$
$2[(3)^2]^{1/4} \times (2)^{1/4}$
$2(9)^{1/4} \times (2)^{1/4}$
$2\sqrt[4]{9} \times (2)^{1/4}$
$2\sqrt[4]{9 \times 2}$
$2\sqrt[4]{18}$ - Ans

Note:
* $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$
* $\sqrt{a} \times \sqrt{a} = (a)^{\frac{1}{2}} = a$
* $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$

Eg → $2\sqrt{3} \times 3\sqrt{27}$
$\Rightarrow 2\sqrt{3} \times 9\sqrt{3}$
$= 18 \times 3 = 54$

Eg → $\frac{2\sqrt{3}}{\sqrt{3}} = 2$

Eg → $\frac{3\sqrt{7}}{7\sqrt{7}} = \frac{3}{7}$ $\left[\frac{3\sqrt{7}}{\sqrt{343}}\right]$

24 May, 22

laws of surds
here a and b are positive Rational number
n → positive integer.
i.e. $n > 1$

1.) $\sqrt[n]{a} \times \sqrt[n]{b} = \sqrt[n]{ab}$
2.) $\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$

3.) $\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a} \rightarrow \sqrt{\sqrt[b]{a}\sqrt{x}} = \sqrt[abc]{x} \rightarrow (x)^{\frac{1}{abc}}$

4.) $\sqrt[n]{a^p} = (a^p)^{1/n} = a^{\frac{p}{n}}$

Properties to solve quardatic surds

- $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$
- $\sqrt{a} \times \sqrt{a} = (a^2)^{1/2} = a^1 = a$
- $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$
- $(\sqrt{a} + \sqrt{b})^2 = (\sqrt{a})^2 + (\sqrt{b})^2 + 2(\sqrt{a})(\sqrt{b})$
  $= a + b + 2\sqrt{ab}$
- $(\sqrt{a} - \sqrt{b})^2 = a + b - 2\sqrt{ab}$
- $(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b \quad ((\sqrt{a})^2 - (\sqrt{b})^2)$
- $(\sqrt{a} + \sqrt{b})(\sqrt{c} + \sqrt{d}) = \sqrt{a}(\sqrt{c} + \sqrt{d}) + \sqrt{b}(\sqrt{c} + \sqrt{d})$
  $= \sqrt{ac} + \sqrt{ad} + \sqrt{bc} + \sqrt{bd}$

Eg - 16.

a.) $(5 + \sqrt{7})(2 + \sqrt{5})$
$\Rightarrow 5(2 + \sqrt{5}) + \sqrt{7}(2 + \sqrt{5})$
$10 + 5\sqrt{5} + 2\sqrt{7} + \sqrt{35}$

b.) $(5 + \sqrt{5})(5 - \sqrt{5})$
$\Rightarrow (5)^2 - (\sqrt{5})^2$
$\Rightarrow 25 - 5$
20

c) $(\sqrt{3} + \sqrt{7})^2$
$\Rightarrow (\sqrt{3})^2 + (\sqrt{7})^3 + 2(\sqrt{3})(\sqrt{7})$
$\Rightarrow 3 + 7 + 2\sqrt{21}$
$\Rightarrow 10 + 2\sqrt{21} = 2(5 + \sqrt{21})$

d.) $(\sqrt{11} - \sqrt{7})(\sqrt{11} + \sqrt{7})$
$\Rightarrow (\sqrt{11})^2 - (\sqrt{7})^2$
$\Rightarrow 11 - 7$
$\Rightarrow 4$

Q.) $2\sqrt{3} - 5\sqrt{12} + 3\sqrt{48}$
$2\sqrt{3} - 10\sqrt{3} + 12\sqrt{3}$
$14\sqrt{3} - 10\sqrt{3}$
$= 4\sqrt{3}$

Q.) $3\sqrt{6} + 2\sqrt{216}$
$3\sqrt{6} + 12\sqrt{6}$
$15\sqrt{6}$

Q.) $\sqrt{32} + \sqrt{48}$
$\sqrt{8} + \sqrt{12}$

$\frac{4\sqrt{2} + 4\sqrt{3}}{2\sqrt{2} + 2\sqrt{3}} = \frac{8}{4} = 2$

Q. $\frac{3\sqrt{12}}{6\sqrt{27}} \Rightarrow \frac{3 \times 2\sqrt{3}}{6 \times 3\sqrt{3}} = \frac{\cancel{6}\sqrt{3}}{\cancel{18}_{3}\sqrt{3}} = \frac{1}{3}$

Q.) $(4\sqrt{3} - 2\sqrt{2})(3\sqrt{2} + 4\sqrt{3})$
$4\sqrt{3}(3\sqrt{2} + 4\sqrt{3}) - 2\sqrt{2}(3\sqrt{2} + 4\sqrt{3})$
$12\sqrt{6} + 48 - 12 - 8\sqrt{6}$
$4\sqrt{6} + 36$

Q1) $(2+\sqrt{3})(3+\sqrt{5})$ — (6)

$\Rightarrow 2(3+\sqrt{5})+\sqrt{3}(3+\sqrt{5})$

$\Rightarrow 6 + 2\sqrt{5} + 3\sqrt{3} + \sqrt{15}$

Q2) $\left(\frac{2}{3}\sqrt{7} - \frac{\sqrt{2}}{2} + 6\sqrt{11}\right) + \left(\frac{1}{3}\sqrt{7} + \frac{3}{2}\sqrt{2} - \sqrt{11}\right)$ — (7)

$\Rightarrow \frac{2}{3}\sqrt{7} - \frac{\sqrt{2}}{2} + 6\sqrt{11} + \frac{\sqrt{7}}{3} + \frac{3\sqrt{2}}{2} - \sqrt{11}$

$\Rightarrow \frac{\cancel{3}\sqrt{7}}{\cancel{3}} + \frac{\cancel{2}\sqrt{2}}{\cancel{2}} + 5\sqrt{11}$

Q8.) $\sqrt{8} + 2\sqrt{32} - \sqrt{50}$ — (8)

$\Rightarrow 2\sqrt{2} + 8\sqrt{2} - 5\sqrt{2}$

$\Rightarrow 10\sqrt{2} - 5\sqrt{2}$

$\Rightarrow 5\sqrt{2}$

Q9.) $\sqrt{12} + \sqrt{192} - \sqrt{147}$ — (9)

$\Rightarrow 2\sqrt{3} + 8\sqrt{3} - 7\sqrt{3}$

$\Rightarrow \cancel{2\sqrt{3}}\ 3\sqrt{3}$

Q10) $4\sqrt{3} + 2\sqrt{75} - 3\sqrt{12}$ — (10)

$\Rightarrow 4\sqrt{3} + 2\times 5\sqrt{3} - 6\sqrt{3}$

$\Rightarrow 4\sqrt{3} + 10\sqrt{3} - 6\sqrt{3}$

$\Rightarrow 14\sqrt{3} - 6\sqrt{3}$

$\Rightarrow 8\sqrt{3}$

Q11. Examine whether following are Rational or irrational.

a.) $\frac{3\sqrt{8}}{\sqrt{2}}$

$\Rightarrow \frac{6\sqrt{\cancel{16}\,2}}{\sqrt{\cancel{2}}} = 6$

b) $\frac{\frac{22}{7}}{\pi}$ = Irrational

c) $(3+\sqrt{2})(2-\sqrt{3})(3-\sqrt{2})(2+\sqrt{3})$

$\Rightarrow (3+\sqrt{2})(3-\sqrt{2})\ (2-\sqrt{3})(2+\sqrt{3})$

$(a+b)\ (a-b) = a^2 - b^2$

$(3)^2 - (\sqrt{2})^2 \times (2)^2 - (\sqrt{3})^2$

$9 - 2 \times 4 - 3$

7×1

$\Rightarrow 7 \Rightarrow$ Rational

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Home work

Eg-13.) Add $2\sqrt{2} + 5\sqrt{3}$ and $\sqrt{2} - 3\sqrt{3}$

sol: $2\sqrt{2} + 5\sqrt{3} + \sqrt{2} - 3\sqrt{2}$

$\Rightarrow 3\sqrt{2} + 2\sqrt{3}$

Eg. 14) Multiply $6\sqrt{5}$ by $2\sqrt{5}$

sol: $6\sqrt{5} \times 2\sqrt{5} = 12 \times 5 = 60$

Eg-15.) Divide $8\sqrt{15}$ by $2\sqrt{3}$

sol: $\frac{8(\sqrt{3}\sqrt{5})}{2\sqrt{3}} = \frac{8\sqrt{3} \times \sqrt{5}}{2\sqrt{3}} = \frac{8\sqrt{5}}{2} = 4\sqrt{5}$ Ans

Ex. 1.5

Q1) Classify the following as rational or irrational

(a.) $2 - \sqrt{5}$

- irrational

b.) $(3 + \sqrt{2}\sqrt{3}) - \sqrt{2}\sqrt{3} = 3$

- Rational

c.) $\frac{2\sqrt{7}}{7\sqrt{7}} = \frac{2}{7}$

- Rational

d.) $\frac{1}{\sqrt{2}}$

- irrational

e.) 2π

- irrational

Spiral

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Q2. simplify

a) $(3 + \sqrt{3})(2 + \sqrt{2})$

sol: $\rightarrow 3(2 + \sqrt{2}) + \sqrt{3}(2 + \sqrt{2})$

$\Rightarrow 6 + 3\sqrt{2} + 2\sqrt{3} + \sqrt{6}$

b.) $(3 + \sqrt{3})(3 - \sqrt{3})$ $[(a+b)(a-b) = a^2 - b^2]$

sol $(3)^2 - (\sqrt{3})^2$

$\Rightarrow 9 - 3$

$\Rightarrow 6$

c) $(\sqrt{5} + \sqrt{2})^2$ $(a^2 + b^2 + 2ab = (a+b)^2)$

$\Rightarrow (\sqrt{5})^2 + (\sqrt{2})^2 + 2(\sqrt{5})(\sqrt{2})$

$\Rightarrow 5 + 2 + 2\sqrt{10}$

$\Rightarrow 7 + 2\sqrt{10}$

d.) $(\sqrt{5})(\sqrt{5} - \sqrt{2})(\sqrt{5} + \sqrt{2})$ $(a^2 - b^2) = (a+b)(a-b)$

$\Rightarrow (\sqrt{5})^2 - (\sqrt{2})^2$

$\Rightarrow 5 - 2$

$\Rightarrow 3$

Spiral